EXAMINATION

Luleå University of Technology

Course: **D7024E**
Course name: **Distributed and Mobile Computing Systems**
Date: **2022-10-24**
Time: **9.00-13.00, online home exam**
Aid: **WWW (no interactive communication with others)**

Teacher on duty (complete telephone number): Olov Schelén, **0920 492237** **olov.schelen@ltu.se**	**Teacher on duty (complete telephone number):**
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Grade scale: **50% grade 3, 75% grade 4 and 90% grade 5**	Note that this grade is for the exam. The exam is supposed to be easy enough so that anyone that followed the course properly should pass it. The grade for the course (including all assessment criteria) may be different from the grade of the exam.
Total number of questions and score: **2 questions, total score 36**	
Other information: This is an individual home exam.	Note: No interactive cooperation is allowed. Information can be searched on the world wide web. Please provide references, when such information is used in you answers.

General instructions

You shall formulate answers with your own words, but for parts of the answers you may cite sources on the web. In any case you must provide the references to your primary sources (e.g., URL). Your **solution must be uploaded in the Canvas D7024E room under Assignments**, at the latest at 13:00 (unless you have gotten approval for extra time). **Please put your name both in the filename and on the first line of your document.**

After examination

Examination results are posted in Canvas within 20 workdays after the examination.

1. **Mixed question on distributed systems (12p)**

   These are multi-choice questions, with three alternatives. Please answer "true", "false" or “depends” to each of the following statements. If your answer is “depends” you must provide a short explanation to get credits. For “true” or “false”, no explanation is required, unless you want to explain what conditions are assumed. If you have identified a corner case and provide a good explanation, we can accept an answer that is different from the expected one. Each question is worth 0.5p.

   a. Distributed systems typically involve multiple nodes with no common clock.

   b. Seamless fail-over typically requires some degree of data replication across nodes, or that they are stateless.

   c. Scalability means that more nodes can be added without affecting performance.

   d. Leader-based consensus solutions can handle large number of nodes and geographical spread.

   e. In a distributed system the nodes typically communicate and coordinate their actions by message passing.

   f. Linear horizontal scalability means that adding more nodes to a distributed system also adds total performance.

   g. It is the idempotence of hashing that makes is suitable for proof of work.

   h. Marshaling (a k a serialization) is part of older packages for communication between heterogeneous computing platforms, but typically not part of more modern approaches.

   i. Kademlia search for a target node is done in O(log n) time when the routing table has been filled, but requires O(n) search to fill the table (where n is the number of nodes).

   j. When there are requirements for immutability of data and idempotence of operations, it becomes really hard to design a decentralized system.

   k. When immediate data consistency is needed (for mutable data), in face of sharding and sufficient replication, the most common approach is to use some kind of index node(s) to guide write operations to storage nodes.

   l. DNS is normally used by SOA and microservices service registries.

   m. SQL databases typically handle and resolve deadlocks due to faulty client implementations, but it hits performance.

n. Blockchain solutions typically provide scale-out transaction processing and storage performance.

o. Leader based global consensus solutions typically assume a non-Byzantine trust model.

p. NAT/firewall traversal by TURN is based on relaying all payload data through an external anchor point.

q. Most pubsub services are decentralized.

r. Decentralized implies distributed, but not the other way around.

s. PGP makes it more difficult to crack encryption by using a random symmetric key (instead of public key) for encrypting the payload.

t. In the message queue paradigm, when there are multiple receivers/subscribers, each message is only delivered to one of them.

u. Big data processing is typically made efficient by avoiding to send data to processing, and instead sending computation to where sharded data is stored.

v. In the Hadoop framework, a combiner function can always be used to decrease the data to be sent to the reducer.

w. Gaming servers and backends are often based on state updates by idempotent messaging.

x. For scalability reasons, online games tend to reduce heavy computation from the clients but may need more network bandwidth.

Note: In the following questions it is important to include sources (e.g., URLs) and that you explain the answer in your own words.

2. Miscellaneous (24p)

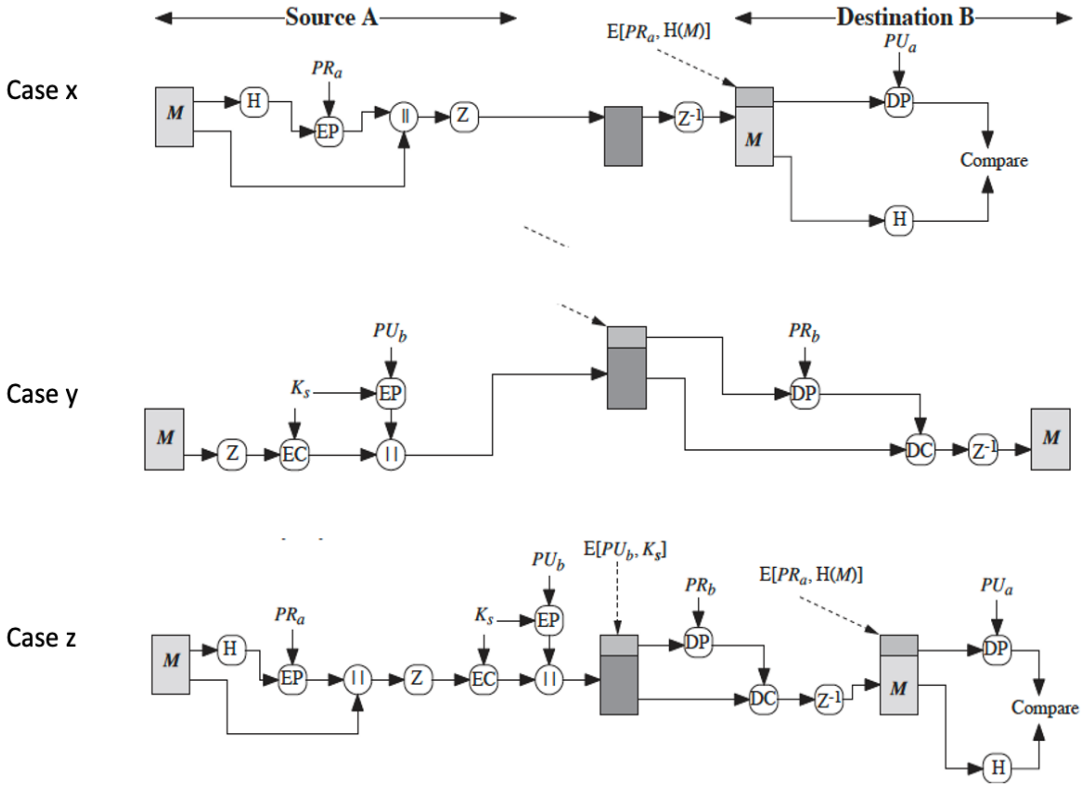
a) In distributed computing systems, explain high availability and fail-over? (2p)

b) Why are so called terms often used in non-byzantine consensus? (2p)

c) In Ethereum, can multiple smart contracts be executed concurrently? Explain. (6p)

d) Explain at least 5 key concepts (implementation approaches) typically used in today's online multiplayer real-time games to provide a good gaming experience with fair synchronization, while keeping small messages and mitigating delay and packet loss (8p)

e) List at least three important dimensions of security, and explain which of those dimensions that are covered in the three cases (x,y,z) of PGP-usage below (6p)



Best Wishes!
Finally, please fill out the **online course evaluation**. An email with a link has been sent to you.